Vacuum cleaners

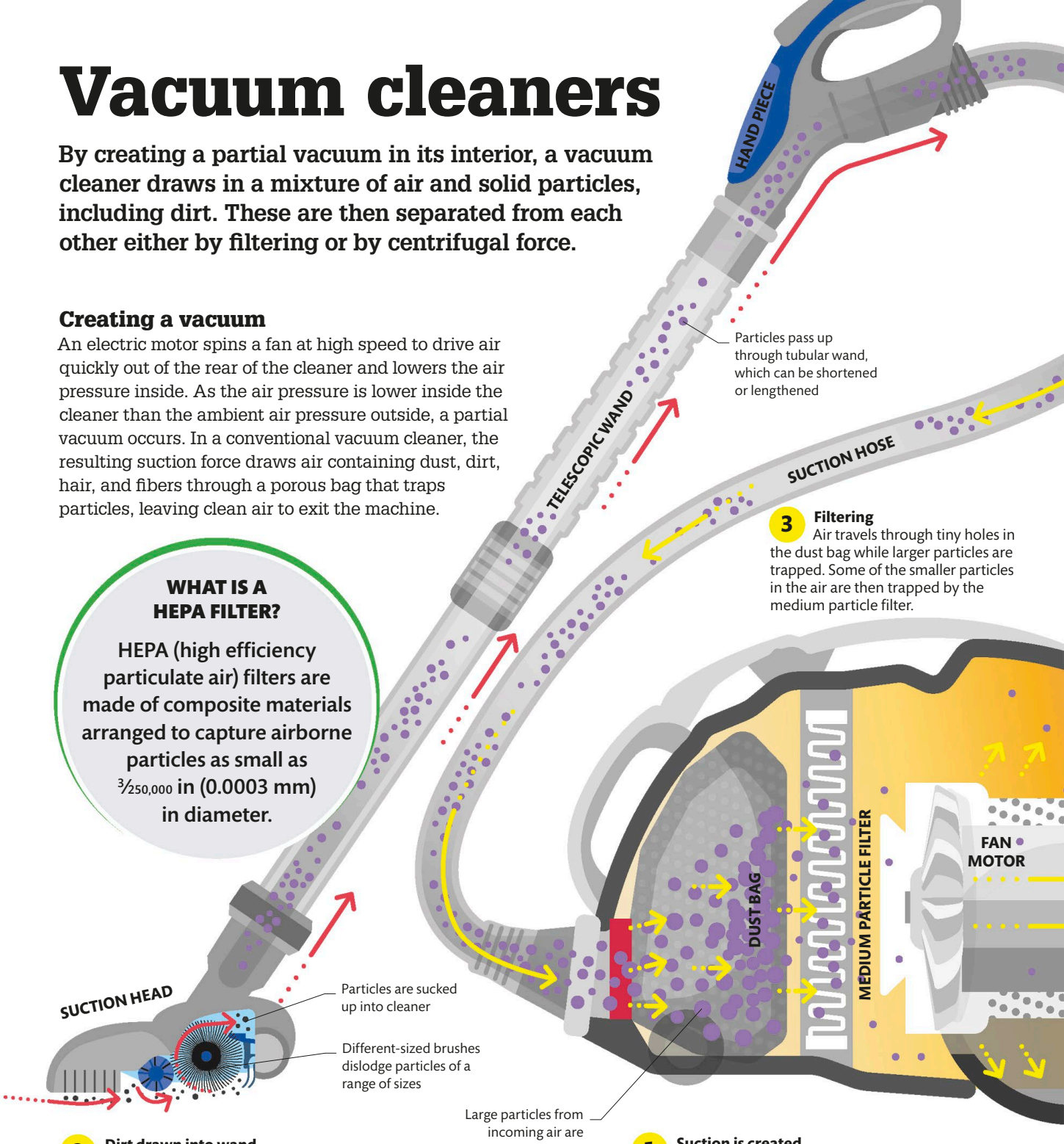
By creating a partial vacuum in its interior, a vacuum cleaner draws in a mixture of air and solid particles, including dirt. These are then separated from each other either by filtering or by centrifugal force.

Creating a vacuum

An electric motor spins a fan at high speed to drive air quickly out of the rear of the cleaner and lowers the air pressure inside. As the air pressure is lower inside the cleaner than the ambient air pressure outside, a partial vacuum occurs. In a conventional vacuum cleaner, the resulting suction force draws air containing dust, dirt, hair, and fibers through a porous bag that traps particles, leaving clean air to exit the machine.

WHAT IS A HEPA FILTER?

HEPA (high efficiency particulate air) filters are made of composite materials arranged to capture airborne particles as small as 3/250,000 in (0.0003 mm) in diameter.



3 Filtering
Air travels through tiny holes in the dust bag while larger particles are trapped. Some of the smaller particles in the air are then trapped by the medium particle filter.

2 Dirt drawn into wand
A series of rotating brushes in the suction head loosen dirt and dust, which is drawn up into the wand and then travels into the cleaner. Most cleaners have a variety of cleaning attachments.

1 Suction is created
The motor spins the fan rapidly to generate suction, which draws air in through the suction head and along the wand and suction hose into the vacuum cleaner body.